

Question 1:

```
//DO NOT MODIFY THIS SECTION
#include <iostream>
using namespace std;
//prototypes HERE

int* create(int);
void analyze(int*, int, int&, int&, float&);

int main() {
    int n, *p, sum, largest;
    float avg;
    cout << "How many numbers?";
    cin >> n;

    //ADD YOUR CODE FROM HERE
    p = create(n);
    analyze(p, n, sum, largest, avg);
    cout << "Sum: " << sum << endl;
    cout << "Largest value: " << largest << endl;
    cout << "Average: " << avg << endl;
    return 0;
}

int* create(int n) {
    int* arr = new int[n];

    // Populate array
    for (int i = 0; i < n; i++) {
        arr[i] = i;
    }
    return arr; // Return memory address of array
}

void analyze(int* arr, const int size, int &s, int &l, float &a) {
    // Set to 0 to avoid garbage value
    s=0;
    l=0;
    a=0;

    // Compute sum
    for (int i = 0; i < size; i++) {
        s += arr[i];
    }

    // Compute largest
    for (int i = 0; i < size; i++) {
        l = arr[0]; // Set to 1st element for the sake of comparison

        if (arr[i] > l) {
            l = arr[i];
        }
    }

    // Compute average
```

```
a = s / size;

// Delete dynamically-allocated array
delete[] arr;
}
```

Function Definition for create:

```
int* create(int n) {
    int* arr = new int[n];

    // Populate array
    for (int i = 0; i < n; i++) {
        arr[i] = i;
    }
    return arr; // Return memory address of array
}
```

Function definition for analyze:

```
void analyze(int* arr, const int size, int &s, int &l, float &a) {  
    // Set to 0 to avoid garbage value  
    s=0;  
    l=0;  
    a=0;  
  
    // Compute sum  
    for (int i = 0; i < size; i++) {  
        s += arr[i];  
    }  
  
    // Compute largest  
    for (int i = 0; i < size; i++) {  
        l = arr[0]; // Set to 1st element for the sake of comparison  
  
        if (arr[i] > l) {  
            l = arr[i];  
        }  
    }  
  
    // Compute average  
    a = s / size;  
  
    // Delete dynamically-allocated array  
    delete[] arr;  
}
```

Function prototypes and main function:

```
→ int* create(int);
→ void analyze(int*, int, int&, int&, float&);

▶ int main() {
    int n, *p, sum, largest;
    float avg;
    cout << "How many numbers?";
    cin >> n;

    //ADD YOUR CODE FROM HERE
    p = create(n);
    analyze(p, size:n, s: [&]sum, l: [&]largest, [&]avg);
    cout << "Sum: " << sum << endl;
    cout << "Largest value: " << largest << endl;
    cout << "Average: " << avg << endl;
    return 0;
}
```

Question 2:

```
#include <iostream>
using namespace std;

int main() {
    int *numbers = nullptr;
    numbers = new int[10];

    for (int i = 0; i < 10; i++) { // Changed <= to <, avoided accessing the
array out of bounds
        *(numbers + i) = i * i;
    }

    for (int i = 0; i < 10; i++) {
        cout << numbers[i] << " ";
    }
    cout << endl;
    delete[] numbers; // Changed delete to delete[], 'numbers' is a
dynamically-allocated array
}
```

```
    return 0;
}
```

Fixes for Question 2:

```
for (int i = 0; i < 10; i++) { // Changed <= to <, avoided accessing the array out of bounds
    *(numbers + i) = i * i;
}
```

```
cout << endl;
delete[] numbers; // Changed delete to delete[], 'numbers' is a dynamically-allocated array
```

Question 1 Output:

```
How many numbers?5

Sum: 10
Largest value: 4
Average: 2

Process finished with exit code 0
```

```
How many numbers?15

Sum: 105
Largest value: 14
Average: 7

Process finished with exit code 0
```

Question 2 Output:

```
0 1 4 9 16 25 36 49 64 81

Process finished with exit code 0
```